

AHMED MUSTAFA

Senior Solutions Architect | Senior Full-Stack Engineer

moonahmed786@gmail.com | +923328371943 | Rawalpindi, Pakistan

<https://www.linkedin.com/in/ahmed-mustafa-b3613754/> | <https://github.com/moonahmed786> | <https://programmersin.com/>

PROFESSIONAL SUMMARY

Solutions Architect & Technical Lead with over a decade of experience, specialising in translating complex business requirements into scalable and robust technical solutions. Expert in MERN (MongoDB, Express, React, Node) and enterprise PHP (Laravel) ecosystems with strong backend architecture skills. Proven experience in designing high-concurrency microservices and delivering Python-based AI solutions (FastAPI/RAG/LLM integrations) across Healthcare, Fintech, E-commerce, and Telecom domains.

PROFESSIONAL EXPERIENCE

Senior Tech Lead & Solutions Architect

Troon Technologies

Jan 2023 – Present | Islamabad

- Led development of a **MERN-based carbon credit platform (RocX)**, boosting system performance through scalable and modular architecture.
- Architected an **AI-powered medical chatbot (AliaMed)** using FastAPI, RAG, and LLMs, greatly improving response accuracy and boosting user engagement.
- Managed integration of **numerous Ruby on Rails applications into Salesforce (CCHMC)**, substantially reducing manual work and improving workflow efficiency.
- Led backend team for **hospital staffing system (Icon Exchange)**, significantly enhancing API performance, achieving massive scalability gains, and implementing robust security improvements.
- Led development of **hospital patient relocation system (PHC)**, significantly improving API speed and reducing latency.
- Delivered enterprise systems across **healthcare, fintech, and AI domains**, improving delivery efficiency and reducing system downtime.
- Specialised in **system architecture, AI integration, and scalable backend development**, consistently improving performance and engineering efficiency across projects.

Senior Software Engineer

Troon Technologies

Jan 2020 – Dec 2022 | Islamabad

- Led development of a **GraphQL API using Laravel** and contributed to the **Vue.js frontend** for a North American healthcare platform (Carehubble), improving **data retrieval efficiency by ~35%** and enabling scalable real-time healthcare workflows •
- Architected and developed a **Laravel + Vue.js cannabis e-commerce platform (MyRemedyApp)** with GraphQL API integration, improving **platform performance by ~30%**, increasing **transaction reliability**, and supporting secure, scalable order management
- Designed and implemented a **nutrition-focused backend system (MyHealthyGut)** using **Yii2 and MySQL**, building robust REST APIs that improved **data processing efficiency by ~40%** and enabled structured health tracking and analytics workflows

Junior Software Engineer

Troon Technologies

Jul 2017 – Dec 2019 | Islamabad

- Developed a **construction analysis and automated reconciliation system (ATS – Allied Technical Service)** using **Yii2, MySQL, and OAuth authentication**, improving **operational accuracy by ~40%** and reducing **manual reconciliation efforts by ~50%** through workflow automation
- Designed and delivered a **PHP-based music learning platform (Cadenza)** featuring lesson management, progress tracking, and structured learning workflows, increasing **user engagement by ~35%** and improving **content management efficiency through scalable platform architecture**

Junior Software Engineer

Green Tech Lab

Aug 2016 – Jul 2017 | Lahore

- Designed and developed a **Laravel-based REST API for a ride-hailing platform (Green Taxi)** featuring geo-based booking, intelligent driver matching, and real-time tracking, improving **trip allocation efficiency by ~35%** and reducing **booking response time by ~30%**
- Implemented scalable backend architecture using **Laravel and MySQL**, increasing **system reliability and concurrent request handling capacity by ~2x** while ensuring stable trip management workflows
- Optimised location-based services and real-time request processing, improving **tracking accuracy by ~40%** and reducing **latency in live ride updates by ~25%**

Full-Stack PHP Developer

PTCL (Pakistan Telecommunication Company Ltd.)

Jan 2015 – Jul 2016 | Lahore

- Developed a **Workforce Management (WFM) HR portal** for call centres using **PHP and MySQL**, automating attendance, shift scheduling, and performance evaluations, resulting in **~50% reduction in manual HR operations** and **~35% improvement in workforce management efficiency**
- Built scalable backend architecture for workforce data management, improving **system reliability and processing efficiency by ~30%** through optimised workflows and automation systems

EDUCATION

Master of Science in Computer Science (M.Sc)

Superior University

Sep 2020 – Dec 2022 | Lahore | GPA: 3.4/4.0

Bachelor of Science in Computer Science (BSCS)

Virtual University of Pakistan

Oct 2012 – Nov 2016 | Lahore | GPA: 2.9/4.0

SKILLS

Languages: PHP, Python, JavaScript (ES6+), TypeScript

Backend Frameworks: Laravel, Yii2, Node.js, NestJS, Express, FastAPI

Frontend: React.js, Next.js, Vue.js

Databases: MySQL, MongoDB, PostgreSQL, Redis, Vector DBs (Pinecone, pgvector)

AI/ML: LLMs, RAG, AI Pipelines, Prompt Engineering, Vector Search

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Docker, CI/CD, GitHub Actions, Jenkins

APIs & Architecture: REST, GraphQL, Microservices, Event-Driven Design, System Design

Tools: Git, JIRA, Postman

LANGUAGES

English (Professional), Urdu (Native), Punjabi (Native), Portuguese (Basic)

KEY PROJECTS

- **AliaMed:** Architected AI medical chatbot (FastAPI, RAG, LLM, Next.js), boosting user engagement by ~40% and improving response accuracy by 35–45% — Healthcare AI
- **RocX:** Led development of MERN-based carbon credit trading platform, improving system performance by ~30% through a scalable architecture — Sustainability
- **CCHMC:** Managed migration and integration of 14+ Rails applications into Salesforce, reducing manual work by ~60% and improving workflow efficiency by ~40% — Enterprise Integration